

CS4740/CS5740
Introduction to Natural Language Processing

Name:

Netid:

Instructions (**READ CAREFULLY****):**

- IF YOU HAVE ANY QUESTIONS ABOUT A QUESTION ON THE TEST, PLEASE JUST INDICATE THIS IN YOUR ANSWER. State any assumptions that you need to make in order to answer the question and then answer the question as best as you can. Only raise your hand to ask us a question if you believe that there is an error in the test.
- Use a **pen or dark-leaded pencil** so that your scanned answer sheets are readable.
- You have **1.5 hours** to complete this exam. The exam is a closed-book / closed-notes / no phones / no cheat sheet / no laptop exam.
- It is **entirely your responsibility** to ensure that your test is **placed directly into the exam collection box** at the front of the classroom at the end of the examination time.

1 N-gram Language models (16 points)

1. (4 points) In one sentence (i.e., your answer should be just one sentence), what is the purpose of smoothing in n-gram language modeling?

Assigns non-zero probabilities to unseen ngrams, hedges against cases where unseen but valid ngrams are assigned 0 probability at test time.

2. (4 points) In one sentence, what is the purpose of replacing rare words in the training corpus with an unknown word identifier?

At test time, we might encounter new words unseen in the training data. Replacing words with $\langle unk \rangle$ during training allows us to learn probabilities for n-grams containing $\langle unk \rangle$ s.

3. The parameters of a 3-gram language model refer to unsmoothed bigram and trigram counts: $\text{count}(w_1, w_2, w_3)$ and $\text{count}(w_1, w_2)$ for all word tuples in a corpus. Suppose we have access to these parameters but the actual corpus C is hidden. **Is each of the following statements true or false? Briefly explain each answer.**

- (a) (4 points) We can apply add-k Laplace smoothing to the 3-gram language model using this count information.

Yes, we can easily determine the vocabulary size $|V|$ from the given information. Then, add-k smoothed probs can be computed as:

$$P(w_3|w_1, w_2) = \frac{\text{count}(w_1, w_2, w_3) + k}{\text{count}(w_1, w_2) + k|V|}$$

- (b) (4 points) We can determine the parameters of a 2-gram language model for C from this information, i.e. $\text{count}(w_1, w_2)$ and $\text{count}(w_1)$.

True. We can determine $\text{count}(w_1)$ by adding up the count of all bigrams that begin with w_1 . We are given $\text{count}(w_1, w_2)$.

2 Text Classification (24 points)

1. (3 points) Consider the task of classifying a given text passage as either Shakespearean or not Shakespearean. Design 3 features that will be useful for this binary classification task. Clearly define how a numerical value for each feature will be extracted from a given text.

Any reasonable feature, like count of “doth” , or even names like “romeo” etc. should be okay.

There should be explicit mapping to a numerical quantity though. (E.g. Feature 1 is count of the number of “doth” in the input.)

2. (3 points) Write the equation for the logistic function $\sigma(z)$.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

3. The rough pseudocode for **binary logistic regression** is given below. Assume $\mathbf{w} = [w_1, w_2 \dots w_k]$ is the weight vector and $\mathbf{f}^j = [f_1^j, f_2^j \dots f_k^j]$ is the feature vector for datapoint j . Everywhere, the **superscript** denotes the index of the datapoint and the **subscript** denotes the index of the weight or feature. We also assume a training dataset $\mathcal{D} = \{(\mathbf{f}^1, y^1), (\mathbf{f}^2, y^2) \dots (\mathbf{f}^N, y^N)\}$, where y^j is the output label.

Algorithm 1 TRAINING BINARY LOGISTIC REGRESSION

1. Initialize $\mathbf{w} = \langle w_1, w_2, w_3 \rangle$
 - for j in $\text{len}(\text{training_data})$:
 2. Compute **Loss for datapoint j , namely L^j**
 3. Compute partial derivative $\frac{\partial L^j}{\partial w_i}$
Hint: for binary logistic regression, $\frac{\partial L^j}{\partial w_i} = f_i^j [\sigma(\sum_i w_i f_i^j) - y^j]$
 4. **Update each w_i in $\mathbf{w}$**
 5. Return $\mathbf{w}$
-

- (a) (4 points) What is the equation for the loss for datapoint j , i.e. L^j .

$$L^j = -\log P(y^j | \mathbf{f}^j).$$

$$\text{where } P(y^j | \mathbf{f}^j) = \frac{1}{1 + e^{-\sum_i w_i f_i^j}} \text{ if } y^j = 1$$

$$\text{else } P(y^j | \mathbf{f}^j) = \frac{1}{1 + e^{\sum_i w_i f_i^j}} \text{ if } y^j = 0$$

- (b) (4 points) What is the update equation in step 4? Use α to denote the the learning rate.

$$w_i = w_i - \alpha \frac{\partial L^j}{\partial w_i}$$

- (c) (10 points) Suppose you have the following training data for binary classification, written in terms of features and labels:

$$\mathbf{f}^1 = \langle 1, 0, -1 \rangle, \quad y^1 = 1$$

$$\mathbf{f}^2 = \langle 1, 1, 1 \rangle, \quad y^2 = 0$$

Run binary logistic regression on this data using stochastic gradient descent. You need to directly follow the pseudocode above. Assume **weights** w_1, w_2, w_3 are initialized to 1, 1, 1 and the **learning rate** is 1. Iterate through the data in the order 1-2 and report the weights w_1, w_2, w_3 after each step. This means you will report 2 values of weight vectors. Show your work.

To simplify calculations, estimate the sigmoid σ as follows:

$$\sigma(z) = \begin{cases} 1/2, & \text{if } z = 0 \\ 1, & \text{if } z > 0 \\ 0, & \text{if } z < 0 \end{cases}$$

$$w^0 = \langle 1, 1, 1 \rangle$$

For data point $j = 1$,

$$\sum_i w_i f_i^j = 0, \quad \sigma(\sum_i w_i f_i^j) = 1/2$$

$$\frac{\partial L^j}{\partial w_1} = 1 \cdot (1/2 - 1) = -1/2$$

$$\frac{\partial L^j}{\partial w_2} = 0 \cdot (1/2 - 1) = 0$$

$$\frac{\partial L^j}{\partial w_3} = -1 \cdot (1/2 - 1) = 1/2$$

Updating weights based on update equation from part (b) and learning rate 1:

$$w_1 = 1 + 1/2 = 3/2, \quad w_2 = 1, \quad w_3 = 1 - 1/2 = 1/2$$

$$w^1 = \langle 3/2, 1, 1/2 \rangle$$

For data point $j = 2$,

$$\sum_i w_i f_i^j > 0, \quad \sigma(\sum_i w_i f_i^j) = 1$$

$$\frac{\partial L^j}{\partial w_1} = 1.(1 - 0) = 1$$

$$\frac{\partial L^j}{\partial w_2} = 1.(1 - 0) = 1$$

$$\frac{\partial L^j}{\partial w_3} = 1.(1 - 0) = 1$$

$$w_1 = 3/2 - 1 = 1/2, \quad w_2 = 1 - 1 = 0, \quad w_3 = 1/2 - 1 = -1/2$$

$$w^2 = < 1/2, 0, -1/2 >$$

3 Word Embeddings (10 points)

1. (4 points) What are two important advantages of dense embeddings (e.g. word2vec) over sparse embeddings (e.g. word co-occurrence or tf-idf)?

Lots of reasonable answers: dense embeddings are have smaller dimensions, less computation to compute similarity, sparse embeddings are sparse :p

2. (6 points) You are given the following data (Table 1) to train a **skip-gram model** with negative sampling. Let C and W be the context and word embedding matrices respectively, where row c_t and w_t denote the context and word vector for token t .

Target word w	Context Word c	label y
jam	peach	+
jam	desk	-
jam	study	-
study	desk	+
study	jam	-

Table 1: Training Data

After training on mutiple epochs on the above training data, which row vectors in C and W might be different from their initial values? Explain why. (Write your answer in terms of c_t and w_t , e.g. use c_{jam} to represent the context vector for “jam”.)

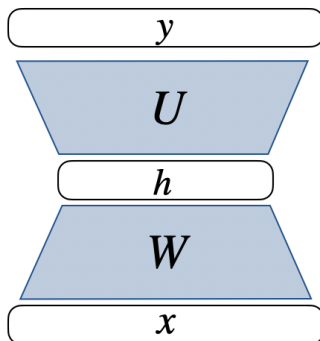
$c_{\text{peach}}, c_{\text{desk}}, c_{\text{study}}, c_{\text{jam}}$

$w_{\text{jam}}, w_{\text{study}}$

4 Feedforward Neural Networks (20 points)

1. (4 points) In one sentence, what is the purpose of a **softmax layer** in a neural network? The purpose of the softmax layer within a neural network is to normalize a vector of reals and ensure that the output values follow the properties of a probability distribution (e.g. all values ≥ 0 , sum to 1).
2. (4 points) In one sentence, what is the purpose of the **activation function** in a neural network?
The purpose of the activation function is to introduce a non-linearity into the network which lets the network learn complex patterns and functions that a linear model alone could not.
3. Consider constructing a feedforward neural network (FFNN) for the task of **trigram language modeling**.
 - (a) (4 points) Provide a diagram of your network. Your network should have the same level of detail as the example below. (This may not be the network you want for the trigram language modeling task!)

B



There are many correct answers depending on whether you use a one-hot encoding and then use embedding weights to convert to an embedding or whether you assume that the input is provide as embedding vectors.

Since we're doing a trigram LM, the input should represent two word tokens.

- (b) (4 points) What is purpose of the input layer of your network? What is its dimension (i.e., size)? Explain your answer.

The input should represent two word tokens. A typical answer would indicate that the dimension is $2 \times \text{length of an embedding vector}$.

Another answer would be $2 \times |V|$ where V is the length of the vocabulary. For this answer to be correct there has to be mention of conversion to embeddings either here or in your diagram.

- (c) (4 points) What is purpose of the output layer of your network? What is its dimension (i.e., size)? Explain your answer.

The output layer needs to show the probability distribution over the vocabulary, indicating the probability that each word will follow the two input tokens. The dimension is $1 \times |V|$.